A3-PS020-TC478_Runtime Rebels

Project Demonstration Results

SRv6 Automated Path Failover Simulation

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1. Problem Statement

Traditional routing protocols experience convergence delays during unexpected link failures, leading to packet loss. The objective of this project is to pre-calculate primary and backup paths to achieve zero-downtime automated failover, ensuring high availability and strict traffic engineering.

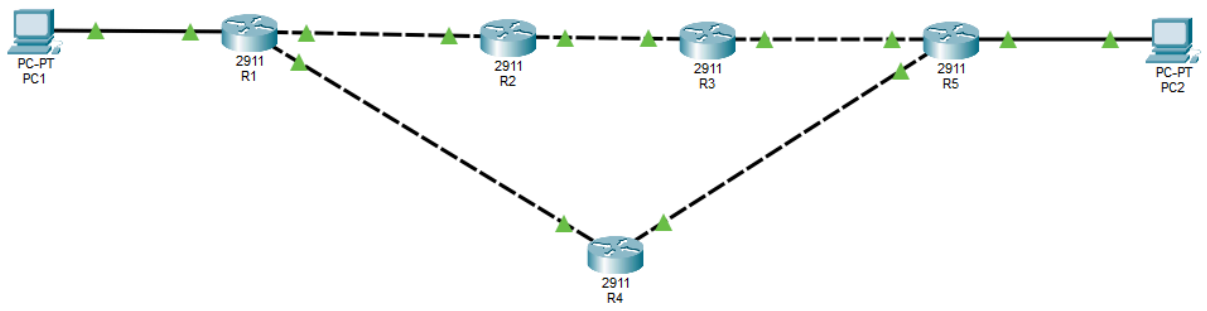
2. Network Topology & Architecture

The simulation is built using a 5-router diamond topology serving as the core transit network connecting PC1 and PC2.

- **Edge Routers:** R1 (Source) and R5 (Destination) handle the path steering logic.
- **Transit Routers:** R2, R3, and R4 act as the provider core forwarding the IPv6 traffic.

Routing Logic:

- Primary path via R2-R3 (Metric 10)
- Backup path via R4 (Metric 20)

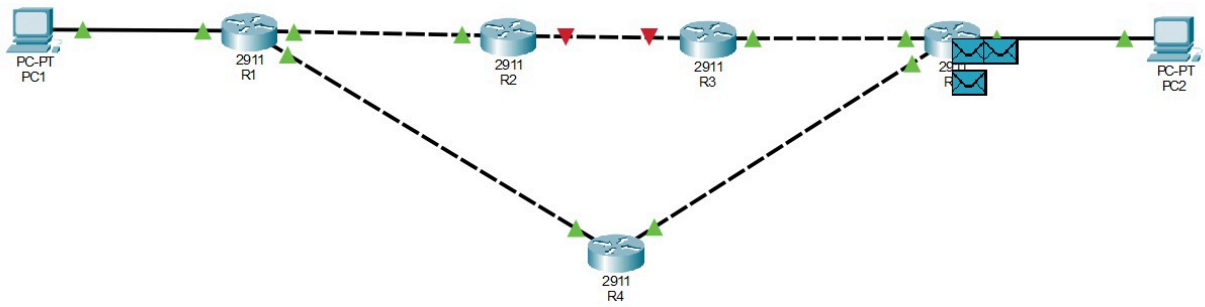


3. Implementation Details

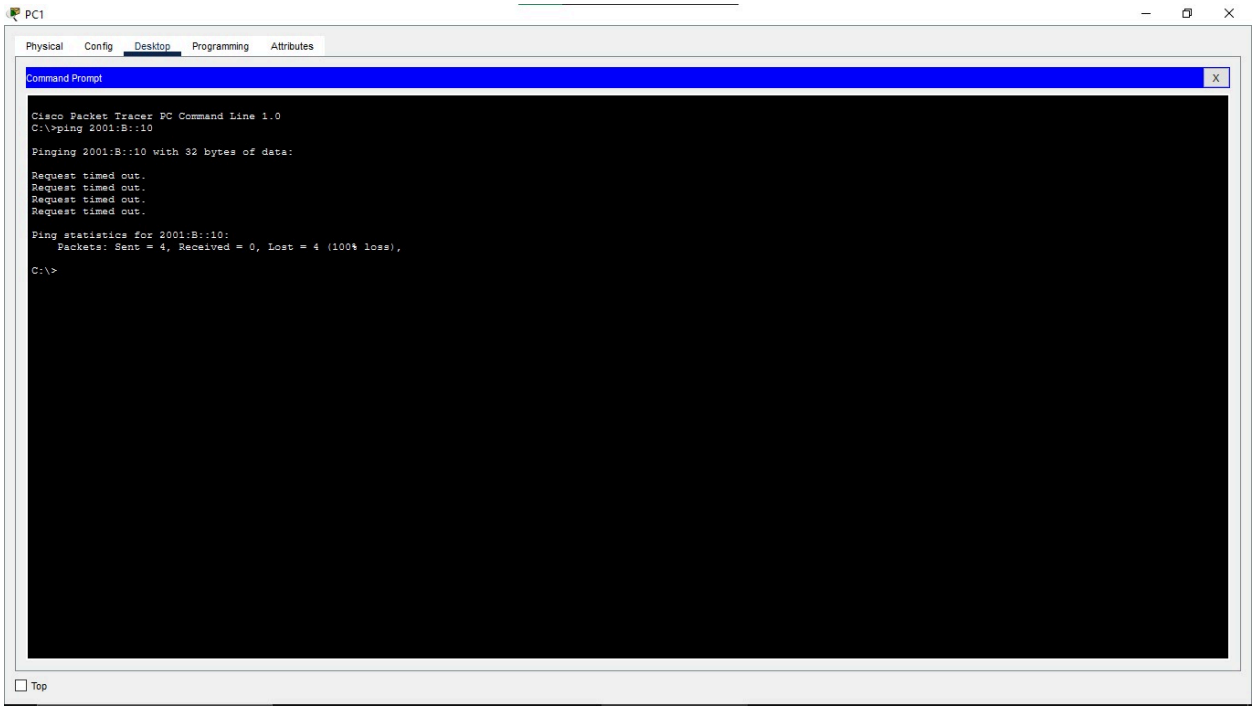
- **Simulation Environment:** Cisco Packet Tracer
- **Protocols:** IPv6, Static Routing, ICMPv6
- **Logic:** Administrative Distance (AD) floating static routes used to simulate explicit SRv6 traffic engineering.
- **Experimental / SDN:** Includes the auto_demo.py Mininet script demonstrating SDN capabilities for native Linux environments.

```
enable
configure terminal
ipv6 unicast-routing
interface g0/0
 ipv6 address 2001:12::2/64
 no shut
interface g0/1
 ipv6 address 2001:23::2/64
 no shut
exit
ipv6 route 2001:B::/64 2001:23::3
ipv6 route 2001:A::/64 2001:12::1
```

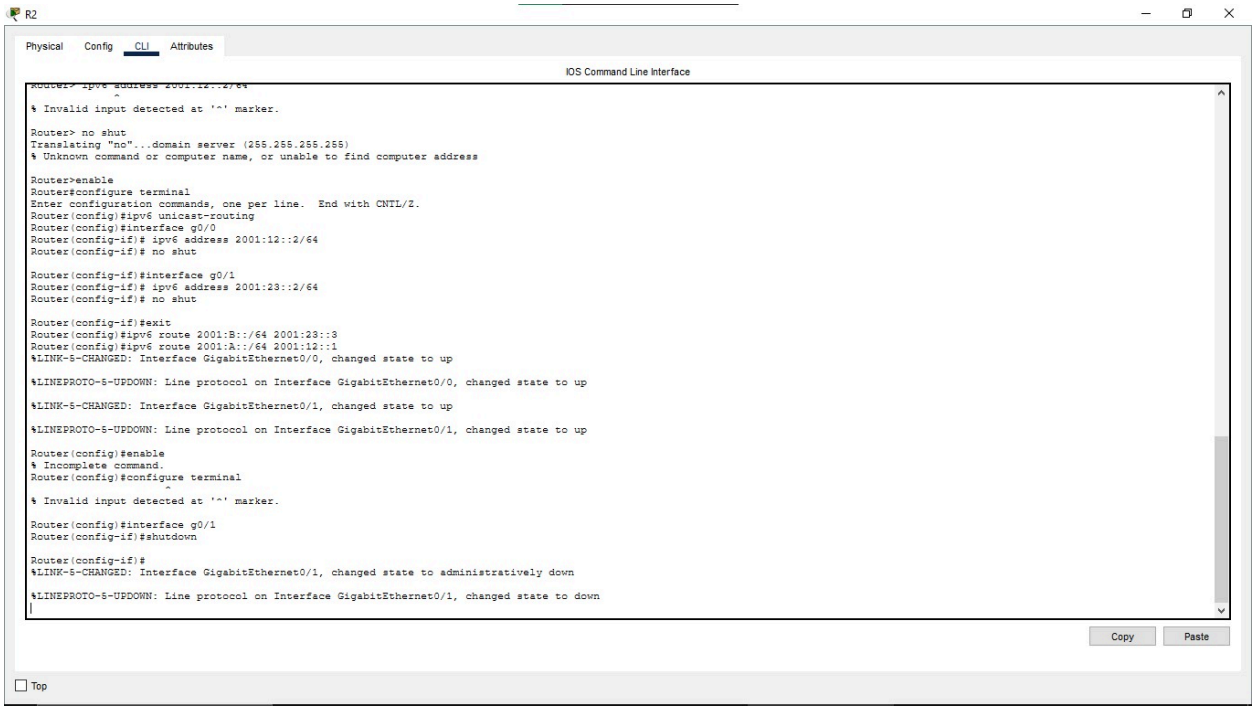
4. Results and Visual Proof



Automated Failover Verification: During a simulated failure of the primary link (R2-R3), the system instantly promotes the AD 20 backup route.



Routing Table Output (Backup Path Active):



5. Future Scope

While the current simulation relies on reactive routing metrics, future iterations will integrate an AI-driven hybrid network diagnostic engine (like NetSage AI) to proactively monitor interface health and steer traffic before a hard failure occurs.

